VINAY KUMAR REDDY CHILAKALA

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SUMMARY

Senior Software Engineer with 9 years of experience architecting peta-byte scale data lakes, large scale distributed systems and zero-ETL pipelines. Proven track record of leading cross-functional teams to build secure CI/CD infrastructure, multi-tenant access controls and backend engineering. Strong background in converting customer usage data into actionable leadership insights. Hands on experience integrating Generative AI to automate complex reporting workflows for executive-level metrics.

EXPERIENCE

Senior Data Engineer, *AWS Migration Services**, Amazon, Boston, MA**.* *Nov, 20 to Present*

Collaborate with cross-functional teams to provide GTM leadership and engineering stakeholders key insights into AWS customers migration journeys. I design, implement and maintain data pipelines/ETL and analytical environment. Engineered end to end system for various tools like *Itemized Credit Statement* which provides key insights into AWS Spend and Credit details by processing petabytes of data daily for 5K+ customers.

Migrations Intelligence Data Lake: A powerful, centralized data repository focused on Customer Migrations Data.

- Technical Lead for organization level initiative involving 5 engineering teams to unify global migration into a data lake solution in AWS. Implemented Zero ETL architecture in data lake to automate data flows from raw data to reporting layer.
- Integrated fine-grained access control using AWS Lake Formation to ensure GDPR and AWS Security compliance.
- Reduced data pipeline latency from 16 to 9 hours and cut Redshift compute costs by $2M annually by optimizing ETL workloads across 30TB of data, utilizing advanced partitioning and data skew mitigation techniques.
- Established Engineering Excellence guidelines for Agentic AI usage data processing infrastructure using CI/CD pipelines with AWS CDK to facilitate multi-region expansion and on-board 15+ teams with no down time.
- Developed comprehensive data observability, alarming and quality frameworks across the MIDL platform, implementing automated anomaly detection and data lineage tracking to ensure high-fidelity data and maintain strict SLAs for executive reporting.
- Architected real-time data ingestion pipelines using AWS Kinesis, streaming high-velocity customer migration events to support high data availability and low-latency downstream analytics.
- On-Boarded ATX Agentic AI for Migrations (VMware, Mainframe, DotNet) global usage data from service teams into MIDL platform and built self-serve reporting layer with Generative AI Capabilities reducing 'Time-to-Insight' for AWS Leadership from days to seconds.
- Accelerated executive decision-making and automated complex data workflows by architecting a multi-agent generative AI ecosystem (Amazon Bedrock, RAG), empowering non-technical leadership to perform natural language querying of customer metrics and automatically generate, visualize, and distribute Weekly Business Reviews via Amazon QuickSight.
    - Managed a team of 2 data analysts and BIE to report latest AWS ATX migrations tools customer adoption metrics directly to AWS S Team.
    - 25+ Interviews for Various Technical Roles
    - 5 yrs consistent high-performance reviews.
    - Ambassador for Data Modelling.

Software Developer, *Indiana Business Research Center, Indianapolis, IN*. *Jan, 18 to Nov, 20*

Developed reusable and domain-driven Data Integration Projects using REST APIs following Client Software Development Life Cycle, performed code reviews, created technical documentation & maintain a high level of code quality and consistency. Modernized State of Indiana websites with new ETL packages & ArcGIS, Developed and Published reports (Analysis & Dashboards) using Tableau, PowerBi for the State of Indiana and IU.

Indiana Data Partnership Program (IDP).

- Designed and developed a Unified Data Model using Master Data Management & Data Virtualization concepts in a team of two developers. Model is trained to consolidate information about organizations from disparate data sources and formats to allow for the dynamic aggregation of data attributes by eliminating redundancy.
- Architect for a scalable big data solution and distributed computation engine built on Hadoop Cluster (setup HDFS, Spark on Yarn, and built a pipeline to import data from various sources to HBase & Hive) to execute Fuzzy Matching using SparkML which improved matching by 75% and speed by 800%.

EDUCATION

M.S. in Computer and Information Science, ***Purdue University, Indianapolis, IN*** [GPA:3.41, Dean's List] *Dec, 19*

Teaching Assistant (TA)

Mentor for undergraduate students for the courses

- Programming Concepts (HTML, CSS, JavaScript).
- Introduction to Databases (SQL, MS-Access).

Thesis Paper: ***Climate Prediction Using Multiple Models.***

Successfully completed and defended thesis in Fall, 19.

Research: ***Multidisciplinary Design Optimization Approach on Climate Prediction.***

Implemented Long Short-Term Memory Convolutional Neural Network (CNN-LSTM) to predict Temperature & Systematically demonstrated how a multi-model approach is superior to any single Model and Reduced RMSE from 1~2% outperforming most Climate Models.

TECHNICAL SKILLS AND CERTIFICATIONS

Computer Languages: *Python, SQL, PySpark, AWS Infrastructure as Code (Cloud Development Kit), TypeScript, Java.*

AWS Services: *Lambda, Redshift, DynamoDB, S3, KMS, Glue, EC2, CloudWatch, Athena, Kinesis, Bedrock.*

Certifications: *Microsoft Certified Solutions Expert (Microsoft Certification ID: 14779288*